Understanding Object Detector Generalization Gaps through the lens of Setting Specificity

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ABSTRACT

Object detectors often perform well in-distribution, yet degrade sharply on a different benchmark. We study cross-dataset object detection (CD-OD) through a lens of setting specificity. To move beyond a simply qualitative grouping, we quantify setting specificity along two independent axes: taxonomic specificity, measured via CLIP-based semantic spread of the label vocabulary, and contextual diversity, measured via a DINOv2 effective-rank and scene-category entropy, validated across six datasets. Through this 'specificity space', we show that taxonomic breadth and contextual diversity are separable dataset properties rather than a single underlying factor. We group benchmarks into setting-agnostic datasets with diverse everyday scenes and setting-specific datasets tied to a narrow environment, and evaluate a standard detector family across all train–test pairs. This reveals a clear structure in CD-OD: transfer is strongly directional, with broad-to-narrow transfer often outperforming the reverse regardless of setting type. The most severe breakdowns predominantly occur when transferring from specific sources to agnostic targets, and persist after open-label alignment, indicating that substantial non-taxonomic visual and capture shift remains in the hardest regimes. To disentangle domain shift from label mismatch, we compare closed-label transfer with an open-label protocol that maps predicted classes to the nearest target label using CLIP similarity. Open-label evaluation yields consistent but bounded gains, and many corrected cases correspond to semantic near-misses supported by the image evidence. We additionally verify that the observed specific→agnostic transfer cliff is not an artifact of a single detector architecture by repeating the Cityscapes↔COCO transfer with RT-DETR and YOLOv11-L, two detectors architecturally distinct from our Faster R-CNN baseline. Overall, we provide a principled framework for evaluating detectors under distribution shift through the lens of setting specificity.

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CCS CONCEPTS

• **Computing methodologies** → **Object detection**; *Computer vision*; *Neural networks*; Machine learning.

KEYWORDS

Object Detection, Domain Generalization, Cross-Dataset Evaluation

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1 INTRODUCTION

Object detection is a safety critical capability in deployed perception systems, and when it fails under rare conditions the consequences can be immediate. In late October 2025, public reporting and released surveillance footage describe a driverless Waymo vehicle in San Francisco fatally injuring a well known neighborhood cat, KitKat, after the animal moved beneath the vehicle while it was stopped and the vehicle then pulled away.[1] This incident is not a basis for attributing failure to any single component, but it highlights a broader deployment reality in which real scenes contain unusual occlusions, rare object trajectories, and operating conditions that are weakly represented in common evaluation suites.

Despite this gap, modern detectors are typically developed and benchmarked within a single dataset ecosystem, which has produced steady progress on standard leaderboards while leaving an important question unresolved: *How does a detector trained on one benchmark behave when evaluated on a different benchmark that reflects different environments and visual statistics while also using a different category vocabulary?* Cross dataset results are also difficult to interpret because two effects are entangled. Performance can drop due to visual distribution shift, but it can also drop due to label taxonomy shift since datasets frequently name, split, or merge concepts differently, so a detector can output a semantically plausible label and still be scored as incorrect under a strict target vocabulary. Transfer tables alone therefore conflate genuine appearance mismatch with evaluation artifacts.

[1] https://www.nytimes.com/2025/12/05/us/waymo-kit-kat-san-francisco.html



(a) (b) (c) (d)

Figure 1: Object detector inference on an image from the COCO dataset. (a) shows the In-domain performance of COCO, whereas (b), (c), and (d) show the Cross-domain performance for a model trained on Cityscapes, Objects365, and BDD100k respectively.

We propose setting specificity as a lens that makes cross dataset results easier to explain, defining it through the intention of the dataset collection. A dataset is setting specific when it targets a core operational setting with repeated scene structure, consistent viewpoints, and a task focused taxonomy, whereas a dataset is setting agnostic when it is collected to cover many settings so capture conditions vary widely and the label space is broader and less concentrated.

This distinction is perceptually salient. A human study of 78 participants, rating visual diversity and judging whether image pairs share a dataset over curated images from COCO [21], Cityscapes [4], Objects365 [31], and BDD100K [38], shows that participants reliably separate homogeneous from diverse collections and recover dataset membership from visual cues alone (Appendix A). This supports the hypothesis that datasets carry stable setting-level signatures, motivating the question of whether detector generalization is similarly organized by setting specificity. We test this with a compact transfer grid covering all combinations of setting types, including within-type transfer where capture geography, weather, sensors, and annotation policy can still shift.

Overall, our contributions are as follows:

- **A quantified, two-axis specificity framework.** We propose a novel framework, quantifying setting specificity as a dataset-level factor. We disentangle specificity into taxonomic specificity (CLIP semantic spread) and contextual diversity (DINOv2 Vendi Score, Places365 scene entropy), across six datasets spanning both axes independently.
- **A setting-aware transfer grid in both closed-label and open-label protocols, separating taxonomy from appearance shift.** We show that train-to-test transfer performance is consistently organized by setting type, and we devise both a closed-label and open-label protocol, to isolate how much of this transfer loss is attributable to label mismatch rather than genuine appearance shift.
- **Cross-architecture check of the transfer cliff.** Under a controlled fine-tuning protocol, we repeat performance transfer with Faster R-CNN, RT-DETR, and YOLOv11-L, showing that the observed specific→agnostic robustness cliff appears across all three detector architectures, and that its severity does not scale with in-domain detector strength.

2 RELATED WORK

2.1 Robustness and Dataset Bias

A growing body of work has shown that strong in-distribution performance does not translate to robustness under naturally occurring distribution shifts. Early studies on dataset bias demonstrated that recognition and detection models often exploit dataset-specific correlations rather than transferable semantics, leading to severe generalization failures across datasets. Torralba and Efros [35] formalized this issue by showing that cross-dataset evaluation reveals substantial bias even when label spaces are aligned.

More recently, robustness benchmarks such as WILDS [14] and COCO-O [23] systematize evaluation under real-world distribution shifts. WILDS covers shifts across time, geography, and data sources, while COCO-O probes appearance and context changes without synthetic perturbations [23].

However, these investigations typically evaluate under controlled shift circumstances, often within a single dataset family. They do not directly address the task of cross-dataset object detection, where models trained on one dataset must generalize to entirely different data collection pipelines, which is more aligned with real-world deployment scenarios.

2.2 Domain Adaptation for Object Detection

Domain adaptation (DA) addresses distribution shift by assuming access to unlabeled or weakly labeled target data. Foundational adversarial alignment methods laid the groundwork for many detector-specific adaptations [2], including strong–weak alignment, selective and categorical alignment, transformer-based adaptation, pixel-level alignment, self-training, and curriculum-based strategies [7, 8, 10, 13, 17–19, 30, 33, 34, 37, 39, 44]. While DA methods demonstrate impressive gains on standard benchmarks such as Sim10k [36]→Cityscapes [4] or Syn2Real [26], they fundamentally rely on target-domain access, either offline or online. As a result, their reported performance conflates robustness with adaptation, making it difficult to assess how detectors behave when deployed on unseen domains without prior exposure. Our benchmark explicitly isolates this missing evaluation regime.

2.3 Domain Generalization and Test-Time Adaptation

Domain generalization (DG) assumes no target-domain access and learns representations invariant across source domains, spanning

feature diversification, alignment regularization, and diffusion-guided designs, with recent variants adding language cues [24]. Test-time adaptation (TTA) permits limited inference-time updates but risks error accumulation and instability. Both typically rely on predefined benchmark splits that can understate real cross-dataset shift; our protocol complements this work by testing frozen detectors across heterogeneous datasets, exposing failure modes adaptation-based evaluation can hide.

2.4 Cross-Domain and Cross-Dataset Object Detection

Cross-domain object detection transfers detectors across datasets with minimal or no target supervision. Benchmarks such as Syn2Real and cross-domain document detection show that domain gaps can be severe even within a single application vertical, motivating alignment, distillation, and disentanglement strategies [3, 9, 11, 15, 22, 28, 32, 40]. Despite this progress, evaluations often rely on a small set of curated source–target pairs and aggregate metrics that mask cross-domain instability, and many protocols implicitly allow some form of adaptation.

In contrast, our benchmark isolates cross-dataset robustness under a unified target-adaptation-free protocol by evaluating frozen detectors across diverse datasets without target-domain training, showing that modern methods remain brittle in the wild despite advances in domain generalization, adaptation, and open-vocabulary learning.

3 METHODOLOGY

3.1 Task Definition and Setting Specificity

Let $\mathcal{D}_s$ and $\mathcal{D}_t$ denote a source and target detection dataset with image spaces $\mathcal{X}_s, \mathcal{X}_t$ and label vocabularies $\mathcal{Y}_s, \mathcal{Y}_t$. In cross-dataset object detection, we train a detector on $\mathcal{D}_s$ and evaluate on $\mathcal{D}_t$, measuring transfer performance under a target evaluation protocol. We report results for all ordered pairs $(\mathcal{D}_s, \mathcal{D}_t)$ in a fixed transfer grid.

We organize datasets by *setting specificity*, defined by the intention of dataset collection. A dataset is *setting specific* if it targets a single operational setting with repeated scene structure, consistent viewpoints, and a task focused taxonomy. A dataset is *setting agnostic* if it is collected to span many settings and scene types, producing broader variation in capture conditions and a less concentrated label space. This induces four transfer regimes given a source type $S \in \{\text{specific}, \text{agnostic}\}$ and target type $T \in \{\text{specific}, \text{agnostic}\}$, namely $S \rightarrow T$.

3.2 Datasets

We evaluate four widely used benchmarks that cover both ends of the setting specificity spectrum. COCO and Objects365 represent setting-agnostic collections with broad object coverage and diverse contexts. Cityscapes and BDD100k represent setting-specific driving collections with structured camera viewpoints and a driving-centric taxonomy.

Across all dataset pairs, we treat taxonomy differences as a first-class confounder during evaluation rather than silently merging concepts. As illustrated in Table 1, COCO and Objects365 can both be classified as setting-agnostic datasets based on the size and diversity of their label spaces. However, Objects365 substantially expands the vocabulary relative to COCO, particularly in fine-grained and rare categories, spanning a wider range of functional and environmental contexts.

Dataset	Images	Classes	Setting
COCO [21]	5,000	80	Agnostic
Objects365 [31]	38,000	365	Agnostic
Cityscapes [4]	500	8	Specific
BDD100k [38]	10,000	10	Specific

Table 1: Dataset statistics for evaluation datasets used in our experiments. *Setting* distinguishes setting-agnostic datasets with generic objects from domain-specific datasets such as driving. The 'Classes' here refer to only the instance-level classes in each dataset.

Cityscapes and BDD100k both have limited and strongly overlapping label-spaces. Cityscapes provides instance-level annotations for only eight object categories, which are all fully contained within the BDD100k label space, which additionally includes only two more labels. All instance-level categories in both datasets are explicitly driving-centric, resulting in a limited, domain-focused ontology. Despite their similarity in label spaces, BDD100k possesses significantly greater diversity in capture conditions when compared with Cityscapes. BDD100k includes both daytime and nighttime scenes, as well as adverse weather conditions such as rain and snow, enabling evaluation under challenging visual conditions.

3.3 Quantifying Setting Specificity

While Section 3.2 groups datasets qualitatively, we additionally quantify setting specificity along two independent, complementary axes: *taxonomic specificity*, capturing how conceptually narrow a dataset's label vocabulary is, and *contextual diversity*, capturing how varied its capture conditions and scene content are.

Taxonomic Specificity (TS).. For each dataset's class vocabulary, we encode class names using the same CLIP ViT-L/14 text encoder used in our open-label protocol (Section 3.6), and compute the mean pairwise cosine distance between class embeddings (*semantic spread*, SS). Narrow, domain-focused vocabularies (e.g., *car, pedestrian, traffic light*) cluster tightly in embedding space; broad, everyday-object vocabularies (e.g., *elephant, umbrella, laptop*) spread widely. We define $\text{TS} = 1 - \text{SS}_{\text{norm}}$ (min-max normalized across datasets), so higher values indicate a narrower vocabulary.

Contextual Diversity (CDS).. Independently, we quantify appearance diversity on a matched sample of N=400 images per dataset using the Vendi Score [5]: the exponentiated Shannon entropy of the normalized eigenvalue spectrum of the covariance matrix of DINOv2 [25] image embeddings, an effective-rank measure of visual diversity. We deliberately use DINOv2 rather than CLIP for this axis, since CLIP's language-aligned embeddings would conflate appearance diversity with the taxonomic axis above. We additionally compute scene-category entropy under a pretrained Places365 [42] classifier. Both signals are bootstrapped across 5 random samples

(mean ± std reported); CDS is their mean after independent min-max normalization.

Dataset	TS ↑	CDS ↑
Cityscapes [4]	1.000	0.115 ± 0.011
BDD100k [38]	0.748	0.144 ± 0.021
VisDrone* [43]	0.458	0.219 ± 0.009
COCO [21]	0.278	1.000 ± 0.005
xView* [16]	0.067	0.239 ± 0.014
Objects365 [31]	0.000	0.962 ± 0.015

Table 2: Taxonomic Specificity (TS) and Contextual Diversity (CDS) scores (min–max normalized) for all six datasets in the specificity space. Higher TS indicates a narrower vocabulary; higher CDS indicates greater appearance diversity. CDS values are reported as mean ± bootstrap std across 5 sampling seeds. TS and CDS are relative coordinates, and the min–max scaling preserves gap ordering. *xView and VisDrone are not included in the final cross-dataset transfer grid (Table 3) as they do not neatly fit into the setting-specific cluster or the setting-agnostic cluster.

Findings. Table 2 and Figure 2 show that TS and CDS agree in ranking our four primary datasets: Cityscapes and BDD100k score high on TS (narrow taxonomies) and low on CDS, while COCO and Objects365 score low on TS (broad taxonomies) and high on CDS, giving convergent-validity support for the qualitative grouping in Section 3.2. However, xView [16] and VisDrone [43] are shown in Figure 2 specifically because they break this diagonal: xView pairs a comparatively broad taxonomy (TS= 0.067, close to Objects365) with contextual diversity closer to the driving-specific cluster (CDS= 0.239), driven by its fixed overhead-satellite viewpoint rather than label-space breadth. VisDrone shows a complementary form of decoupling: an intermediate, moderately narrow taxonomy (TS= 0.458, between BDD100k and COCO) paired with contextual diversity (CDS= 0.219) similarly close to the driving-specific cluster. Both aerial datasets therefore reach comparably low contextual diversity from very different points on the taxonomic axis, reinforcing that the two axes vary independently rather than as a single latent factor. This decoupling is not an edge case but a design consequence: taxonomic breadth is set by annotation policy, while contextual diversity is set by capture protocol, and a dataset can sit anywhere in the resulting two-dimensional space. This indicates that "setting specificity" must not conflate two separable dataset properties – what a dataset labels, and how it was captured.

Key finding

Taxonomic specificity and contextual diversity are empirically separable. xView and VisDrone reach driving-like contextual diversity from opposite ends of the taxonomic axis, so setting specificity is inherently two-dimensional and cannot be captured by a single score.

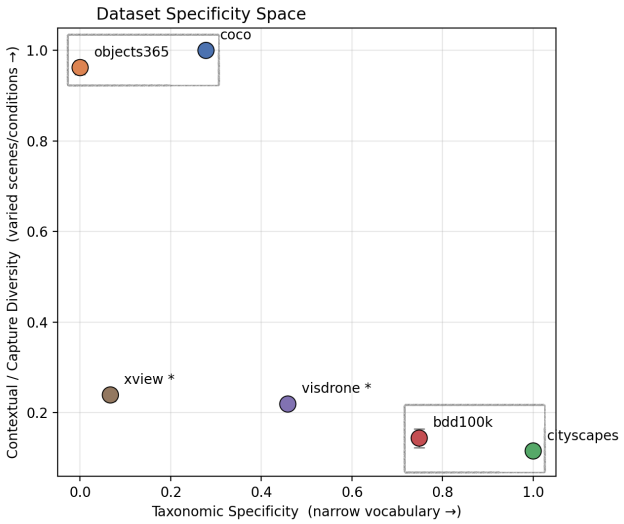


Figure 2: Dataset specificity space (Taxonomic Specificity vs. Contextual Diversity). COCO/Objects365 and Cityscapes/BDD100k occupy distinct clusters in opposite corners; xView [16] and VisDrone [43] occupy an intermediate region that decouples the two axes. (* indicates datasets not included in the cross-dataset transfer experiments)

3.4 Training protocol

To isolate dataset effects, we fix the training recipe across all source datasets, train one model per benchmark, and run all training and inference through a single codebase to avoid toolkit-dependent variation. All models are trained only on the source dataset, using no target-domain images, annotations, or parameter updates; target class names are used only for evaluation-time label alignment (Section 3.6). Faster R-CNN [29] with a ResNet50 FPN [6][20] backbone was used as a common architecture due to its stability and widespread adoption in detection benchmarks, its explicit decoupling of localization and classification, and its balanced model capacity that enables fair and controlled comparison across datasets without confounding architectural effects.

To verify that observed transfer patterns are not an artifact of a single architecture, we additionally conduct a controlled cross-architecture check across three detectors: Faster R-CNN, RT-DETR [41], and YOLOv11-L [12]. For each architecture, we begin from a COCO-pretrained checkpoint and evaluate it directly on COCO and Cityscapes (columns COCO→COCO and COCO→City in Table 6); we then fine-tune a copy of that checkpoint on Cityscapes' training split (2,975 images) and evaluate the fine-tuned model on Cityscapes and COCO (columns City→City and City→COCO). All four evaluations are restricted to the same 7 classes shared by COCO and Cityscapes. This differs from the from-scratch, single-source training used for Table 3's Faster R-CNN results; it isolates architecture effects under a shared, controlled fine-tuning protocol rather than reproducing that table's numbers. This tests whether the specific→agnostic robustness cliff observed in Section 4 is also observed across other architecture families under a common fine-tuning setup.

3.5 Implementation Details

All experiments are implemented in PyTorch using the MMDetection framework [1] with a Faster R-CNN ResNet-50 FPN backbone and a standard 1× learning-rate schedule. Evaluations are performed on the validation splits of COCO, Cityscapes, Objects365v2, and BDD100K. In ablations, we compute semantic similarity using CLIP ViT-L/14[27] text embeddings with raw label strings as prompts. We apply open-label remapping after NMS, leaving all boxes and IoU matching unchanged.Experiments are conducted on an NVIDIA RTX A5000 GPU (24 GB VRAM).

3.6 Evaluation protocols and metrics

We report a transfer grid over all source–target pairs, within and across setting types, using COCO-style AP averaged over IoU thresholds 0.50–0.95 together with size-specific AP.

Closed-label evaluation. Scoring is restricted to the one-to-one intersection of shared labels between source and target, aligned through explicit mapping files with case-insensitive normalization and no semantic collapsing; predictions outside the shared set are filtered before scoring.

Open-label evaluation. Since closed-label scoring penalizes semantically plausible predictions that fail exact matching, the open-label protocol reassigns each predicted label $\hat{y}$ to its nearest target label $c \in \mathcal{Y}_t$ when the best text similarity exceeds a threshold τ, and leaves it unchanged otherwise; localization is untouched throughout.

Semantic near-miss diagnostics. To confirm that open-label gains reflect meaningful near-misses rather than arbitrary remapping, we characterize each label mismatch by three CLIP-based quantities: text-to-text similarity $s_{tt}(\hat{y}, y) = \cos(e_T(\hat{y}), e_T(y))$; the rank of the ground-truth label under the prediction, $\text{rank}_{\text{GT}|\hat{y}} = 1 + \sum_{c \in \mathcal{Y}_t} \mathbb{1}[\cos(e_T(\hat{y}), e_T(c)) > \cos(e_T(\hat{y}), e_T(y))]$; and a region-level margin $\Delta_{it} = s_{it}(r, y) - s_{it}(r, \hat{y})$ with $s_{it}(r, c) = \cos(e_I(r), e_T(c))$ the image–text similarity on region crop r. A mismatch counts as a near-miss when $s_{tt} \geq \tau$ and $\text{rank}_{\text{GT}|\hat{y}} \leq K$.

4 QUANTITATIVE RESULTS

Table 3 reports closed-label cross-dataset transfer, where evaluation is restricted to the shared label intersection for each train–test pair. Even under this conservative protocol, cross-dataset performance drops sharply relative to in-domain scores, indicating that the dominant failure mode is not only taxonomy mismatch but also appearance and capture bias. Because the shared label intersection differs across pairs (e.g., the BDD–Objects365 intersection is much smaller than the COCO–Objects365 intersection, as discussed below), the row and column averages below should be read as descriptive summaries over this pairwise-intersection grid rather than as a fully controlled, intersection-free measure of dataset-level transferability.

Across sources, Objects365 achieves the highest mean AP in this pairwise-intersection grid (0.213) despite only a moderate in-domain score (0.198), making it the strongest source among those tested here. In particular, Objects365→COCO reaches 0.286 and Objects365→Cityscapes reaches 0.268, both substantially higher than any transfer originating from the driving-specific sources. In contrast, Cityscapes has the lowest mean AP as a source in this grid (0.115), with especially poor transfer to COCO (0.015) and Objects365 (0.020), despite achieving the strongest in-domain score of any dataset in the grid (0.402). This indicates that strong in-domain performance on a narrow, structured domain does not translate into transferable representations.

Finally, apparent target difficulty varies substantially: averaged over all sources, Objects365 has the lowest mean AP as a target in this grid (0.092), while Cityscapes has the highest (0.287). This pattern is consistent with Cityscapes and BDD sharing a nearly identical, narrow driving taxonomy, so that any driving-trained source aligns well with Cityscapes' label space, whereas Objects365's large vocabulary and high scene variation are difficult to hit precisely for detectors trained on other, narrower datasets.

This asymmetry is not uniform across pairs. Taking Objects365 as the test set, BDD (0.102) and COCO (0.046) as sources differ sharply despite COCO sharing Objects365's setting type; but with Objects365 as the source, COCO (0.286) and BDD (0.103) as targets show the opposite gap. The BDD–Objects365 pair is thus close to symmetric (0.102 vs. 0.103), while COCO–Objects365 is highly asymmetric (0.046 vs. 0.286).

One plausible explanation is that the BDD–Objects365 label intersection is restricted to a few canonical, high-frequency categories (*car*, *person*, *truck*) that a narrow driving detector localizes reliably, while the larger COCO–Objects365 intersection makes transfer more dependent on training-set scale and density—consistent with Objects365's larger, denser training set (Section 5). Similarly, findings in Section 5 show that Objects365 → COCO predictions achieve high IoU but with hallucinated detections. We flag this as a target for future class-level analysis rather than a settled explanation.

4.1 Open-Label Cross-Dataset Evaluation

Closed-label transfer penalizes semantically plausible predictions that fail exact name matching across taxonomies. To estimate how much transfer loss is attributable to this effect, Table 4 reports open-label evaluation, where predicted class names are mapped to the nearest target label by CLIP text similarity (with $\tau = 0.6$), leaving localization unchanged. Open-label matching yields consistent but bounded gains on all off-diagonal entries. The improvements are largest for transfers between the two agnostic datasets and for transfers from the broader vocabularies into driving datasets. For example, Objects365→COCO improves from 0.286 to 0.322 (+0.036), and COCO→Objects365 improves from 0.046 to 0.082 (+0.036). COCO→Cityscapes improves from 0.206 to 0.235 (+0.029), while COCO→BDD improves from 0.131 to 0.160 (+0.029). These gains indicate that a meaningful portion of cross-dataset error is due to label mismatch and naming granularity rather than incorrect boxes.

At the same time, open-label evaluation does not close the robustness gap. Several transfers remain extremely low even after semantic remapping, such as Cityscapes→COCO (0.030) and BDD→COCO (0.038). This residual gap points to harder transfer phenomena, including viewpoint and context shift, long-tail frequency changes, and localization degradation that cannot be repaired by relaxing label equivalence.

Table 3: Cross-dataset object detection performance (mAP@0.50–0.95) under closed-label evaluation. Rows indicate the test (target) dataset; columns indicate the train (source) dataset.

	Train				
Test	COCO[21]	Cityscapes[4]	Objects365[31]	BDD[38]	**Avg.**
COCO[21]	0.376	0.015	0.286	0.019	0.174
Cityscapes[4]	0.206	0.402	0.268	0.275	0.287
Objects365[31]	0.046	0.020	0.198	0.102	0.092
BDD[38]	0.131	0.025	0.103	0.298	0.139
Avg.	0.189	0.115	0.213	0.174	–

Table 4: Cross-dataset object detection performance (mAP0.50–0.95) under open-label evaluation (cosine similarity ≥ 0.6 to nearest test label). Rows indicate the test (target) dataset; columns indicate the train (source) dataset. Diagonal entries are unchanged. Red subscripts denote gains over closed-label evaluation.

	Train				
Test	COCO[21]	Cityscapes[4]	Objects365[31]	BDD[38]	**Avg.**
COCO[21]	0.376	$0.030_{+0.015}$	$0.322_{+0.036}$	$0.038_{+0.019}$	0.192
Cityscapes[4]	$0.235_{+0.029}$	0.402	$0.296_{+0.028}$	$0.287_{+0.012}$	0.305
Objects365[31]	$0.082_{+0.036}$	$0.032_{+0.012}$	0.198	$0.124_{+0.022}$	0.109
BDD[38]	$0.160_{+0.029}$	$0.040_{+0.015}$	$0.132_{+0.029}$	0.298	0.158
Avg.	0.213	0.126	0.237	0.187	–

Table 5: Open-label diagnostics using CLIP. We analyze matched detections where the box is correct (IoU matched) but the label may differ. Near-miss rate is the fraction of label mismatches that satisfy $s_{tt} \geq 0.6$ and $\text{rank}_{GT|\hat{y}} \leq 5$.

Train→Test	ΔmAP	**Near-miss** ↑	$\mathbb{E}[s_{tt}]$ ↑	Med. Rank GT$\|\hat{y}$ ↓	Top-5 ↑	$\mathbb{E}[\Delta_{it}]$ (% $\Delta_{it} > 0$)
Objects365 [31] → BDD [38]	+0.029	41%	0.73	2	88%	+0.018 (62%)
Objects365 [31] → City [4]	+0.028	33%	0.71	3	81%	+0.012 (58%)
COCO [21] → Objects365 [31]	+0.036	18%	0.69	4	64%	+0.006 (55%)
Objects365 [31] → COCO [21]	+0.036	17%	0.70	4	66%	+0.005 (54%)
City [4] → COCO [21]	+0.015	4%	0.58	14	21%	−0.002 (46%)

4.2 CLIP-Based Ablation: Diagnosing Taxonomy Mismatch vs. Semantic Confusion

Table 5 analyzes IoU-matched detections whose labels disagree under closed-label scoring. We report the near-miss rate, defined as the fraction of mismatches that remain semantically close to the ground-truth label under the target vocabulary (text–text similarity above threshold and the ground-truth ranked in the top-K candidates). We also summarize the strength of semantic proximity through the mean text similarity, the median ground-truth rank, and the top-5 inclusion rate.

Transfers that benefit from open-label evaluation exhibit substantially higher near-miss rates and stronger semantic proximity. For instance, Objects365→BDD shows a 41% near-miss rate with high mean similarity (0.73), a low median rank (2), and 88% top-5 inclusion, indicating that many label "errors" are in fact semantically adjacent predictions. Objects365→Cityscapes shows a similar pattern with 33% near-misses and 81% top-5 inclusion. By contrast, Cityscapes→COCO has a near-miss rate of only 4%, low similarity (0.58), and poor rank statistics, suggesting that its remaining errors are dominated by genuine semantic confusion and domain shift rather than taxonomy mismatch.

4.3 Robustness Across Detector Architectures

Table 6 confirms that the qualitative asymmetry identified in Section 4 – specific→agnostic transfer being harder than agnostic→specific transfer – persists for all three fine-tuned detectors. Normalizing each model's cross-dataset score by its own in-domain score, City→COCO retention is 40.9% for Faster R-CNN (0.106/0.259), 35.4% for RT-DETR (0.142/0.401), and 26.9% for YOLOv11-L (0.090/0.334), while COCO→City retention is more similar across models, at 54.8%, 47.4%, and 45.9% respectively. The resulting asymmetry gap (COCO→City retention minus City→COCO retention) is therefore largest for YOLOv11-L (~19 points) and comparable for the other two models.

Notably, relative retention under this controlled fine-tuning protocol does not track architecture family in a simple way: the two-stage Faster R-CNN retains proportionally *more* performance in both transfer directions than either single-stage detector, despite RT-DETR [41] and YOLOv11-L [12] being substantially stronger in-domain detectors overall (COCO→COCO of 0.561 and 0.542 vs. 0.376 for Faster R-CNN). This indicates that stronger absolute in-domain performance does not automatically confer better relative robustness to cross-dataset transfer.

Separately, comparing Faster R-CNN's two training regimes across Tables 3 and 6 is informative in its own right, though the two are not a controlled comparison of the same variable. Fine-tuning from a COCO-pretrained checkpoint (this table) retains substantially more source-domain (COCO) performance than training from scratch on Cityscapes alone (Table 3): City→COCO is 0.106 (40.9% relative retention) for the fine-tuned model versus 0.015 (3.7%) for the from-scratch model, at some cost in peak in-domain Cityscapes score (0.259 vs. 0.402). This suggests the severe robustness cliff under from-scratch training reflects the *absence* of general-purpose pretrained features rather than an architectural limitation of Faster R-CNN, demonstrating the benefit of pretrained initialization, not catastrophic forgetting in the technical sense.

Model	COCO→COCO	COCO→City	City→City	City→COCO
Faster R-CNN	0.376	0.206	0.259	0.106
RT-DETR	0.561	0.266	0.401	0.142
YOLOv11-L	0.542	0.249	0.334	0.090

Table 6: Closed-label mAP@0.50–0.95 for three detectors under a matched cross-architecture protocol, restricted to the 7 classes shared by COCO and Cityscapes (excluding *rider*, absent from COCO). COCO→COCO and COCO→City report the original COCO-pretrained checkpoint evaluated directly on each dataset; City→City and City→COCO report the same checkpoint after fine-tuning on Cityscapes' training split, evaluated on both datasets. All three models start from COCO-pretrained weights, so these Faster R-CNN values are not directly comparable to the from-scratch, single-source Faster R-CNN results in Table 3.

5 QUALITATIVE RESULTS

Figure 3 presents qualitative results across different domain transfer settings. Firstly, Figs. 3b, j, and 3p confirm that in-domain transfers (Objects365→Objects365, COCO→COCO, and Cityscapes→Cityscapes) consistently yield the most reliable predictions. Figs. 3e, f, and m demonstrate strong generalization, where predictions align closely with ground truth for cross-domain transfers like COCO→Objects365 and Objects365→Cityscapes, indicating robust cross-domain feature learning. Similarly, Figs. 3c and d show successful transfer to urban scenes, where major objects and pedestrians are localized accurately in Cityscapes-like environments; alongside this, Figs. 3q and s illustrate strong cross-dataset transfer to daylight city scenes in BDD. However, performance degrades in challenging conditions such as nighttime (Fig. 3t) or glare (Fig. 3r) scenes. Although models trained on Cityscapes or COCO are attuned to standard layouts, their limited diversity compared to BDD100k prevents reliable generalization to these extreme cases. Figs. 3a, h, and o illustrate failure modes where overlapping objects or background clutter lead to incorrect predictions. Figs. 3i and l further highlight severe degradation under low-light conditions and large domain shifts, resulting in missed detections or misclassifications. Finally, while Fig. 3g achieves high IoU for Objects365→COCO, it suffers from excessive hallucinated detections, reflecting a precision–recall trade-off.

6 DISCUSSION

Setting specificity decouples into separable axes. The four primary datasets fall along a single diagonal in two clusters (Figure 2), yet xView [16] and VisDrone [43] sit off it, reaching low contextual diversity from very different taxonomic breadths through an overhead-satellite and an aerial viewpoint respectively. The two axes respond to distinct design choices, what a dataset labels versus how it was captured, and do not reduce to a shared latent factor. Taking the gap as source minus target, ΔTS and ΔCDS correlate with closed-label AP over the 12 off-diagonal pairs at $\rho = -0.70$ and +0.64: sources with a narrower vocabulary or lower contextual diversity than their target transfer worse; unsigned gap magnitudes correlate only weakly (Spearman $\rho = -0.16, -0.33$). With only four datasets, we treat this as a directional trend rather than a causal effect.

Transfer is directional and collapses out of narrow domains. Many pairs in Table 3 are strongly asymmetric, and the asymmetry has a consistent shape: transfer from setting-specific sources into setting-agnostic targets is the predominant failure regime, staying low even after open-label scoring (Table 4). This locates the collapse in appearance, viewpoint, and context shift rather than naming; the near-miss diagnostics (Table 5) confirm that label mismatch, while real, accounts for only a bounded share of the gap. Strong in-domain performance on a narrow, structured setting thus offers little guarantee once the target scene distribution widens.

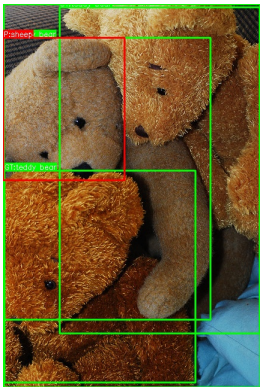
Key finding

One-way, curated source→target pairs can hide the harder direction and overstate deployment readiness; full transfer grids are needed.

The asymmetry is architecture-general; training regime drives its severity. Under a matched fine-tuning protocol (Table 6), specific →agnostic transfer remains the harder direction for Faster R-CNN [29], RT-DETR [41], and YOLOv11-L [12] alike, marking a dataset-level effect. Severity is independent of in-domain strength, as the weakest detector retains the most under transfer and the strongest the least. The controlling factor is the presence of general-purpose pretrained features, evident in the gap between Faster R-CNN's fine-tuned and from-scratch variants.

Key finding

In-domain accuracy is not a reliable proxy for cross-dataset robustness; pretrained initialization matters more than detector family.



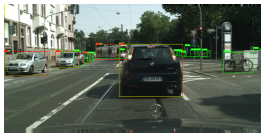
(a) O365→COCO



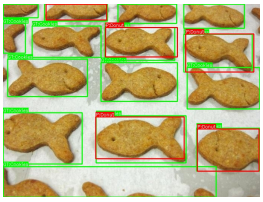
(b) O365→O365



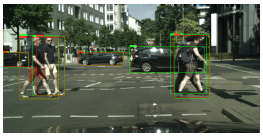
(c) O365→City



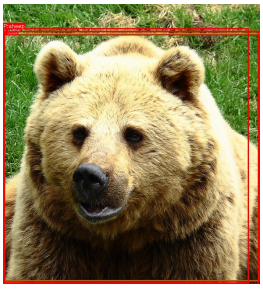
(d) COCO→City



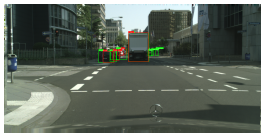
(e) COCO→O365



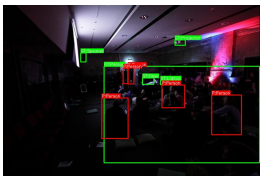
(f) O365→City



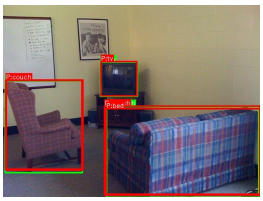
(g) O365→COCO



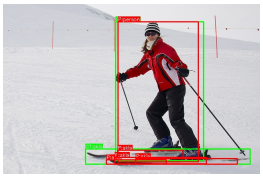
(h) COCO→City



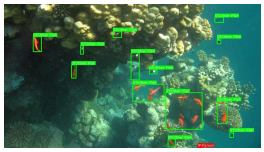
(i) COCO→O365



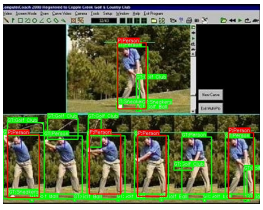
(j) COCO→COCO



(k) COCO→COCO



(l) City→O365



(m) City→O365



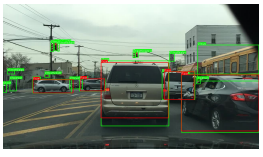
(n) City→COCO



(o) City→COCO



(p) City→City



(q) COCO→BDD



(r) COCO→BDD



(s) City→BDD



(t) City→BDD

Figure 3: In-domain and cross-domain qualitative results. City refers to images from Cityscapes, and O365 refers to images from the Objects365 dataset. X→Y denotes a model trained on dataset X and evaluated on dataset Y.

7 CONCLUSION

We identify setting specificity as a useful lens for understanding cross-dataset object detection, and quantify it along two independent axes, taxonomic specificity and contextual diversity, showing across six datasets that Objects365/COCO and Cityscapes/BDD occupy opposed clusters in this space. We then introduce a unified protocol that evaluates a frozen detector across all train–test pairs under both closed-label and open-label scoring, isolating semantic misalignment from visual domain shift. Our findings show that transfer is strongly directional, with the most severe degradation when transferring from narrow, setting-specific sources to diverse, setting-agnostic targets, a robustness cliff that persists after open-label alignment and suggests that appearance and capture shift matter beyond label mismatch. Overall, we provide a principled framework for evaluating detectors under distribution shift, and future work can pursue domain-adaptation strategies against the shifts identified here.

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A HUMAN PERCEPTION STUDY

To verify that setting specificity is perceptually salient, we conducted a study with 78 participants using curated images from COCO [21], Cityscapes [4], Objects365 [31], and BDD100K [38]. Each participant rated four 5-image sets on a 1–10 Likert scale (1 = very uniform, 10 = very diverse) and answered five Yes/No questions on whether a pair of images came from the same dataset collection. Table 7 summarizes results by response type.

Likert ratings were strongly polarized across the four sets (median 4, IQR $[2, 8]$): 65.4% of responses fell within ±1 of the per-question mode despite only 35.3% exact agreement, indicating consistent relative judgments even without a shared absolute scale. On the same/different-dataset task, participants were highly accurate (84.0% overall), and markedly more so when the correct answer was "different" (89.7%) than "same" (80.3%). These results indicate that datasets exhibit stable setting-level signatures recoverable from visual cues alone.

Table 7: Human study summary.

Likert responses

Metric	Value
Mean	4.83
Median [IQR]	4 [2, 8]
Accuracy (exact GT)	35.3%
Agreement (within ±1 of GT)	65.4%

Yes/No responses

Metric	Value
Accuracy	84.0%
Accuracy (GT = Yes)	80.3%
Accuracy (GT = No)	89.7%
Per-question accuracy range	66.2–98.7%